

IP addressing & Subnetting

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANS:

Class	IP Range	Example Device/Service
A	1.0.0.0 – 126.255.255.255	Large company network
B	128.0.0.0 – 191.255.255.255	University network
C	192.0.0.0 – 223.255.255.255	Home WiFi router
D	224.0.0.0 – 239.255.255.255	Video streaming multicast
E	240.0.0.0 – 255.255.255.255	Research and testing

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.

ANS :

Given:

- IP Address = 192.168.10.45
- Subnet Mask = 255.255.255.0 (/24)

Step 1: Network Address

- Set host bits to 0.
- Network Address = 192.168.10.0

Step 2: Broadcast Address

- Set host bits to 1.
- Broadcast Address = 192.168.10.255

Step 3: Valid Host Range

- First Host = 192.168.10.1
- Last Host = 192.168.10.254

- Final Answer

Item	Address
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Host	192.168.10.1
Last Host	192.168.10.254

- 3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

ANS :

Using 10.0.0.0/24

Split into four /26 subnets:

Group	Network Address	Subnet Mask
Delivery Partners	10.0.0.0/26	255.255.255.192
Restaurants	10.0.0.64/26	255.255.255.192
Customers	10.0.0.128/26	255.255.255.192
Admins	10.0.0.192/26	255.255.255.192

Each subnet supports 62 hosts.

4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint : Show your calculation steps and final answers.

ANS :

Given:

- Network = 172.16.0.0/16
- Need at least 20 subnets

Step 1: Borrow Bits

Need 20 subnets.

- $2^4 = 16$ (not enough)
- $2^5 = 32$ (enough)

Borrow 5 bits.

Step 2: New Subnet Mask

Original = /16

New = /16 + 5 = /21

Subnet Mask = 255.255.248.0

Step 3: Hosts Per Subnet

Remaining host bits = $32 - 21 = 11$

Hosts = $2^{11} - 2$

= $2048 - 2$

= 2046 hosts

- Final Answer

Item	Value
New Subnet Mask	255.255.248.0 (/21)
Number of Subnets	32
Hosts per Subnet	2046

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

ANS :

The IP address 192.168.1.300 is invalid because each number (octet) in an IPv4 address must be between 0 and 255.

Here, 300 is greater than 255, so the device cannot use this address.

Valid examples in the same subnet:

- 192.168.1.10
- 192.168.1.50
- 192.168.1.200

A valid IP in this subnet should look like:

192.168.1.x (where x is between 1 and 254).